Question 1:

To be answered in **(20) Minutes** [ ] / **13 Marks**

#	Questions	T/F
1	Given two heaps with n elements each, it is possible to construct a single heap comprising all 2n elements in O(n) time.	
2	Running merge sort on an array of size n which is already correctly sorted takes O(n) time	
3	On a connected, directed graph with only positive edge weights, Bellman-Ford runs asymptotically as fast as Dijkstra	
4	If $f(n) = O(g(n))$ and $g(n) = O(h(n))$, then $h(n) = \Omega(f(n))$	
5	The following array is a max heap: [10, 3, 5, 1, 4, 2].	
6	If $f(n) = 5n\log n$ and $g(n) = n \log 5n$ then $f = \Theta(g)$	
7	Building a heap with n elements can always be done in O(n log n) time	
8	If $f(n) = O(g(n))$ and $g(n) = O(f(n))$ then $f(n) = g(n)$	
9	Divide & Conquer algorithms solving approach is top down.	
10	Every problem in NP can be solved in exponential time	
11	Breadth first Search (BFS) graph search algorithm uses QUEUE data structure	
12	Problems that can be solved in polynomial time are known as decision	
13	Fractional knapsack algorithm is a greedy algorithm that items can be divided to have a maximum benefit.	

Question 2:

To be answered in (20) Minutes [8] / 8 Marks

1. **Solve the recursion $T(n)= 2T(n-1)$ and $T(0)=1$ using iteration method. [3 marks]**

2. **Can we solve the previous recurrence in Q 1 using the master theorem method? Justify your claim? [1 mark]**

3. **Revised the Merge sort algorithm to divide the n-element sequence to be sorted into four (instead of two) subsequences of n/4 elements each. Give pseud code and complexity? [4 marks]**

Question 3: To be answered in **(20) Minutes** [ ] / 7Marks

4. Consider the following array to answer each question: [7 marks]

4	10	3	5	1

a) Give the heap tree [1 mark]

b) If *Build-Max-Heap* algorithm is applied to the array, draw the steps of created heap. [2 marks]

c) Draw the array after applying **one** step of *Heapsort* algorithm using *Max* heap (i.e., after popping off 10 from the root.) [1 mark]

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d) If *Min* heap is built instead of *Max* heap, what is the value of the root? [1 mark]

e) Draw the final value of the array after applying *Heapsort* algorithm when *Min* heap is used instead of *Max* heap. [2 marks]

Question 4: *NP-complete and Graph* To be answered in (20) Minutes [ ] / 12 Marks

Q1: list the name of 3 algorithms hav time complexity (2^n) [3 marks]

Q2: Using dynamic programming of 0-1 knapsack algorithm below, the capacity is 5 and you have 3 items, Compute the value of an optimal solution, and list items of this solution. [2 marks]

Item 1: Value = 6, Weight = 1
Item 2: Value = 10, Weight = 2
Item 3: Value = 12, Weight = 3

		Capacity					
		0	1	2	3	4	5
Item	0	0	0	0	0	0	0
	1	0	6	6	6	6	6
	2	0	6	10	16	16	16
	3	0	6	10	16	18	22

Time complexity of dynamic programming of 0-1 knapsack algorithm is: [1 mark]

Q3: Fill in the missing parts of the following pseudocode for Prim's algorithm to build Minimum Spanning Trees. [6 marks]

Student ID:

```
MST-PRIM(G, w, r)
 1  for each u ∈ G.V
 2        u.key = ∞
 3        u.π = NIL
 4  r.key = 0
 5  .....
 6  while .....
 7        u = .....
 8        for each v ∈ G.Adj[u]
 9              if ..... and w(u, v) < v.key
10                    v.π = u
11                    v.key = w(u, v)
```

The running time for the algorithm above in following lines:

Line 1-5 :.....

Line 6:..

Line 7: ....

Line 8: .....

The total running time of the algorithm:

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